

All
About
Pizza!



PIZZA

SALES ANALYSIS
USING SQL & Excel

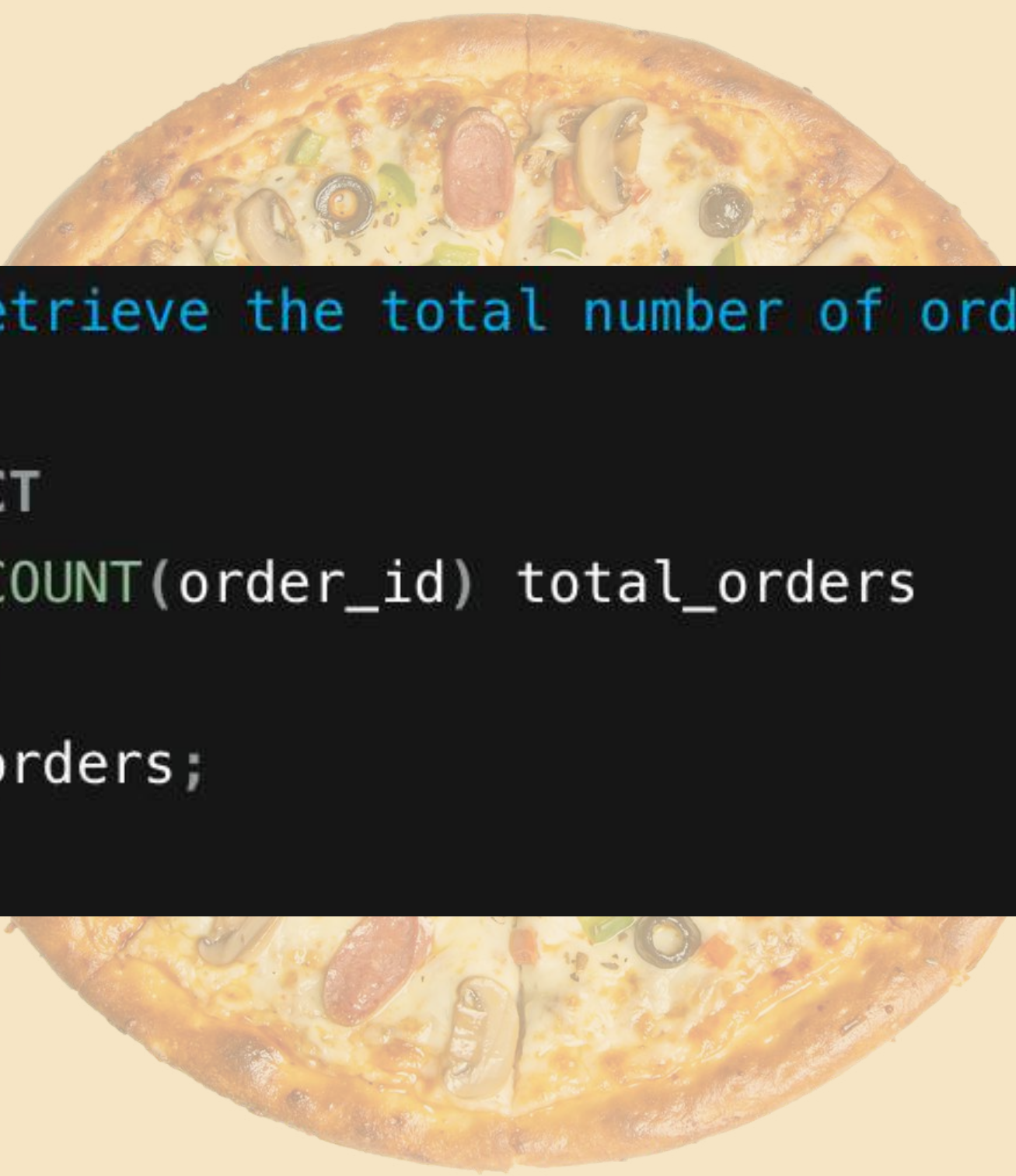
ABOUT ME

Hi, I'm Ayush, an aspiring Data Analyst / Data Scientist passionate about turning raw data into meaningful business insights. I enjoy working with SQL, Excel, Python, Statistics, and Data Visualization to solve real-world business problems.

BASIC QUESTIONS

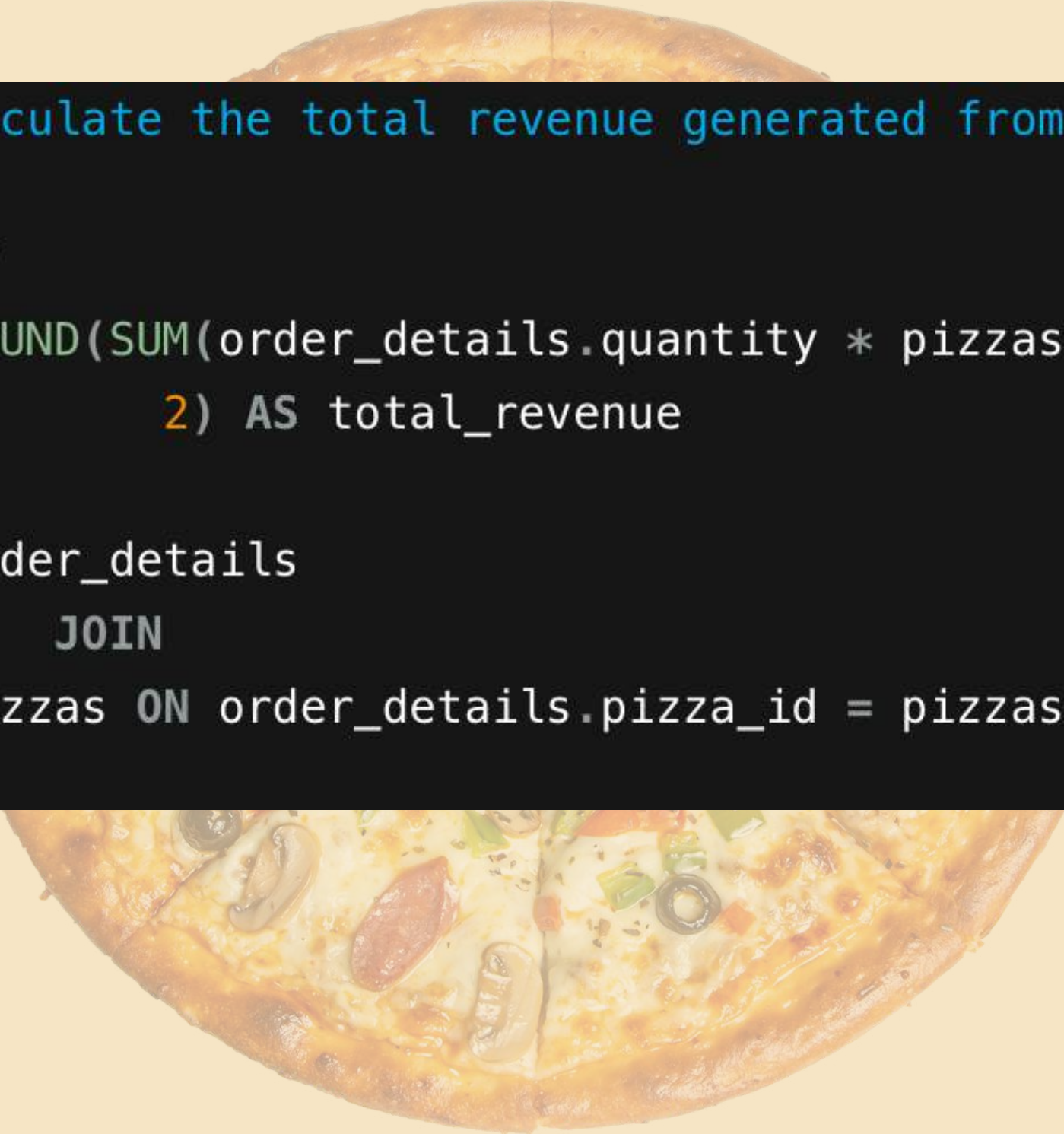


TOTAL NUMBER OF ORDERS PLACED



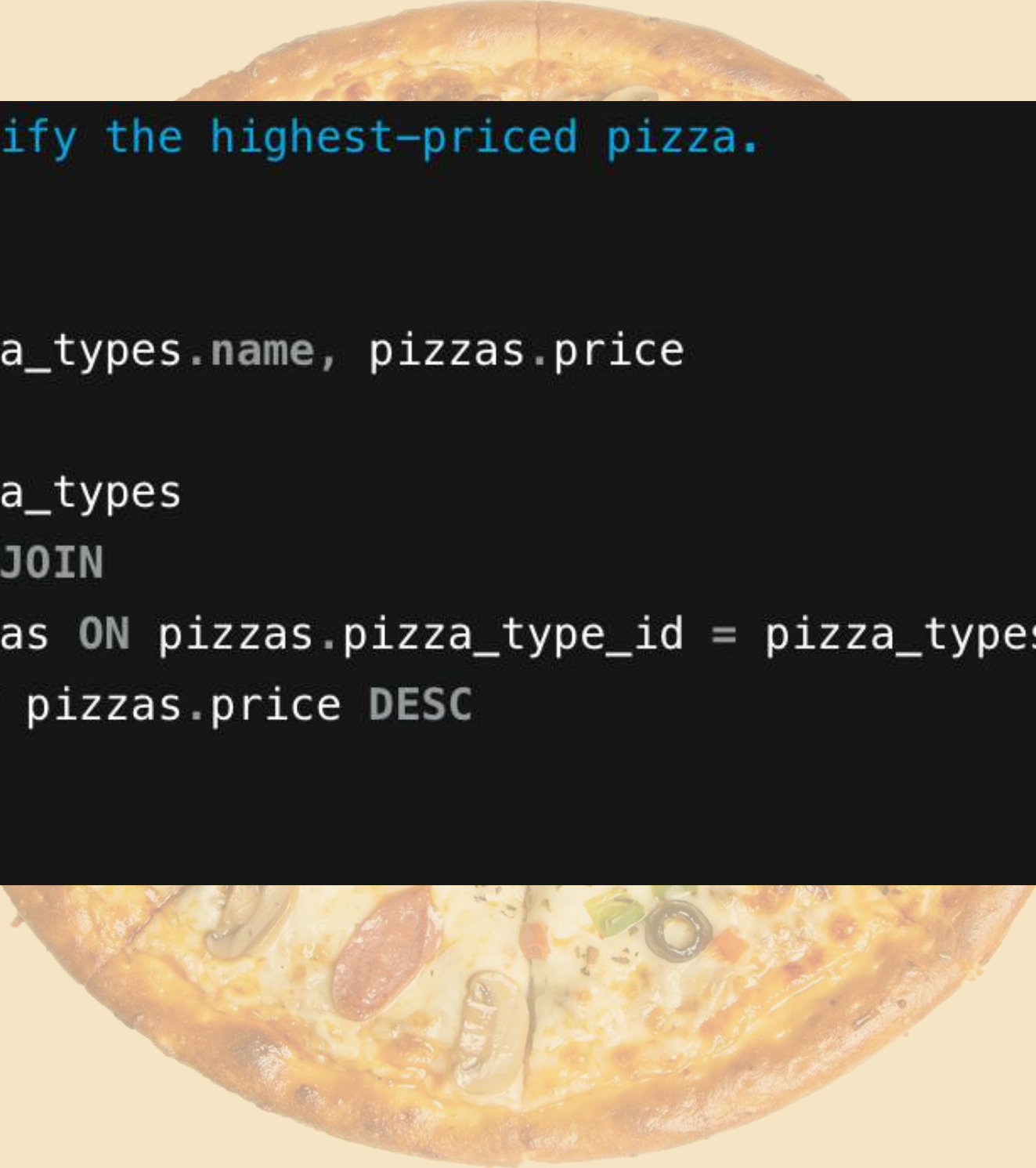
```
1  -- Retrieve the total number of orders placed.  
2  
3  * SELECT  
4      COUNT(order_id) total_orders  
5  FROM  
6      orders;  
7
```

TOTAL REVENUE GENERATED



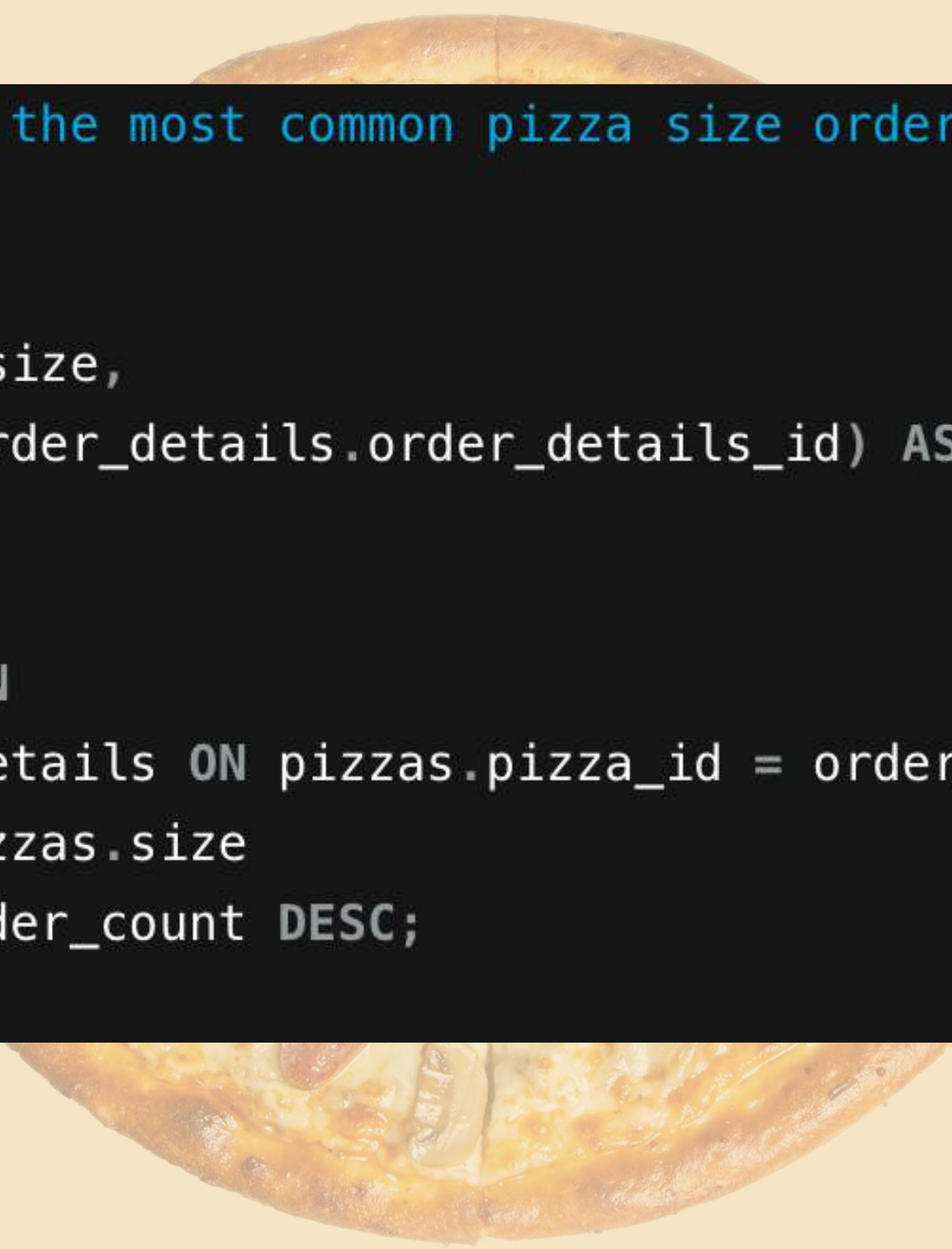
```
1  -- Calculate the total revenue generated from pizza sales.
2
3  SELECT
4  ⊖    ROUND(SUM(order_details.quantity * pizzas.price),
5           2) AS total_revenue
6  FROM
7      order_details
8      JOIN
9      pizzas ON order_details.pizza_id = pizzas.pizza_id;
10
```

HIGHEST-PRICED PIZZA



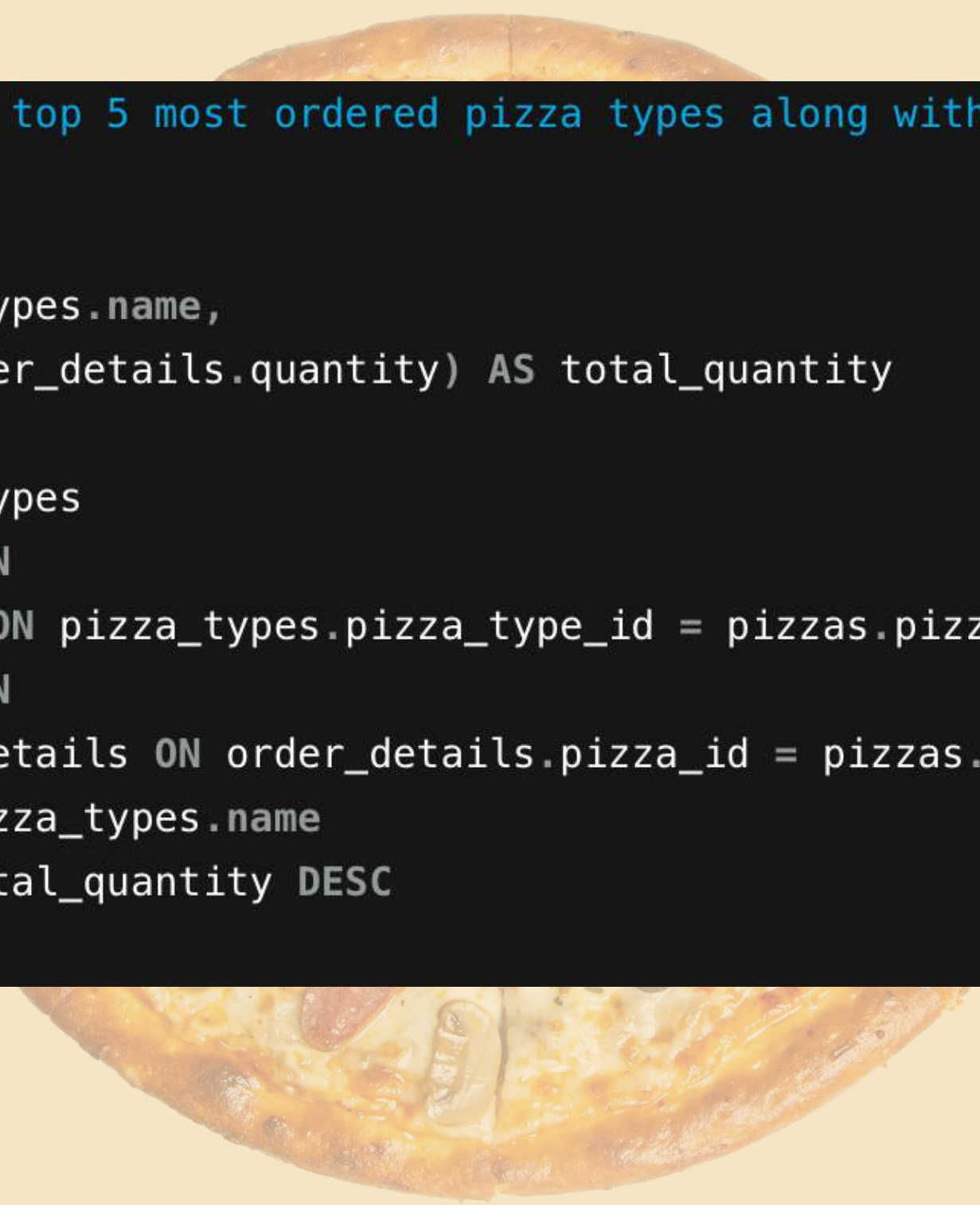
```
1  -- Identify the highest-priced pizza.
2
3  * SELECT
4      pizza_types.name, pizzas.price
5  FROM
6      pizza_types
7      JOIN
8      pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
9  ORDER BY pizzas.price DESC
10 LIMIT 1;
11
```


MOST COMMON PIZZA SIZE



```
1  -- Identify the most common pizza size ordered.
2
3  SELECT
4      pizzas.size,
5      COUNT(order_details.order_details_id) AS order_count
6  FROM
7      pizzas
8      JOIN
9      order_details ON pizzas.pizza_id = order_details.pizza_id
10 GROUP BY pizzas.size
11 ORDER BY order_count DESC;
12
```

TOP 5 MOST ORDERED PIZZA TYPES

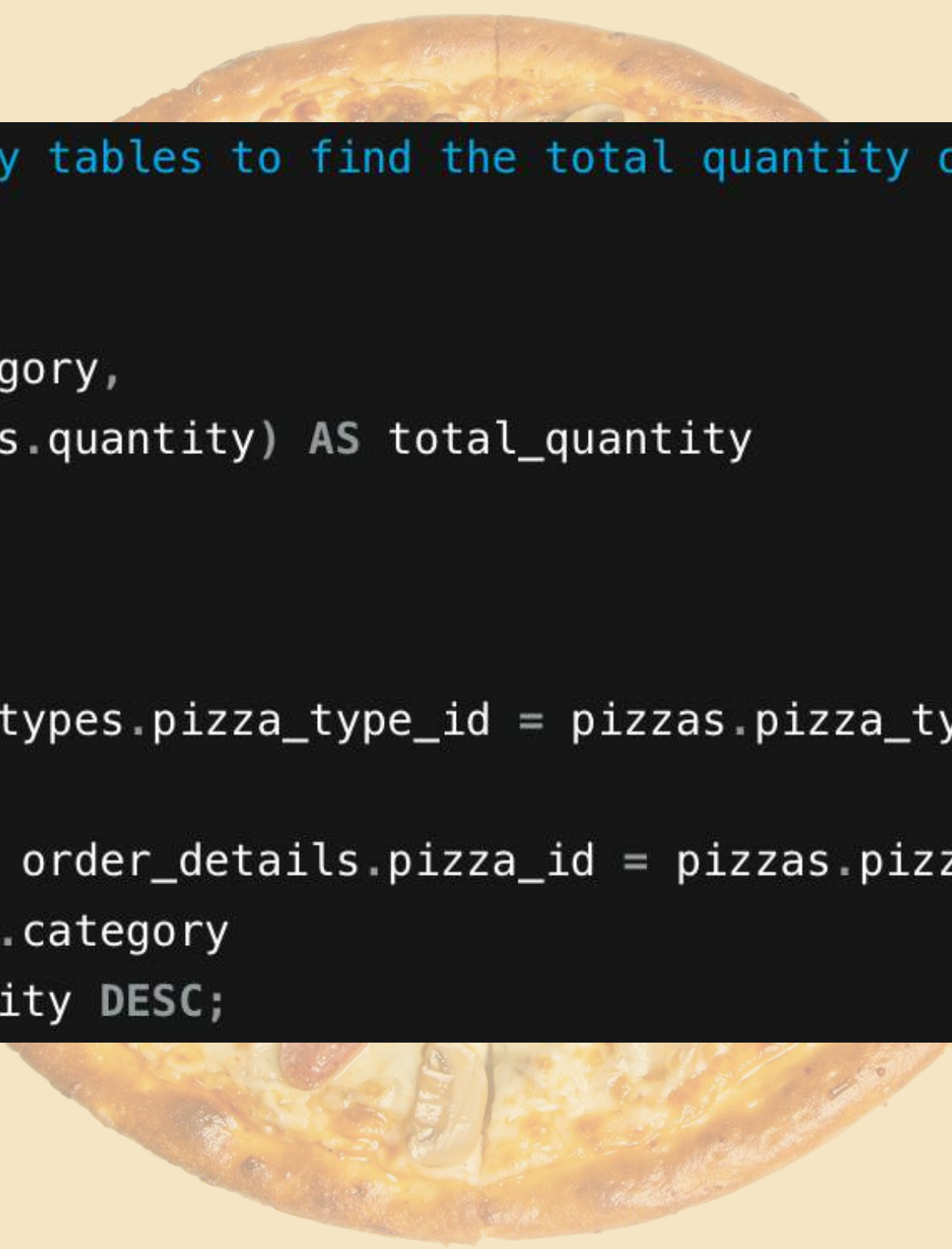


```
1  -- List the top 5 most ordered pizza types along with their quantities.
2
3  SELECT
4      pizza_types.name,
5      SUM(order_details.quantity) AS total_quantity
6  FROM
7      pizza_types
8      JOIN
9      pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
10     JOIN
11     order_details ON order_details.pizza_id = pizzas.pizza_id
12 GROUP BY pizza_types.name
13 ORDER BY total_quantity DESC
14 LIMIT 5;
```


INTERMEDIATE QUESTIONS

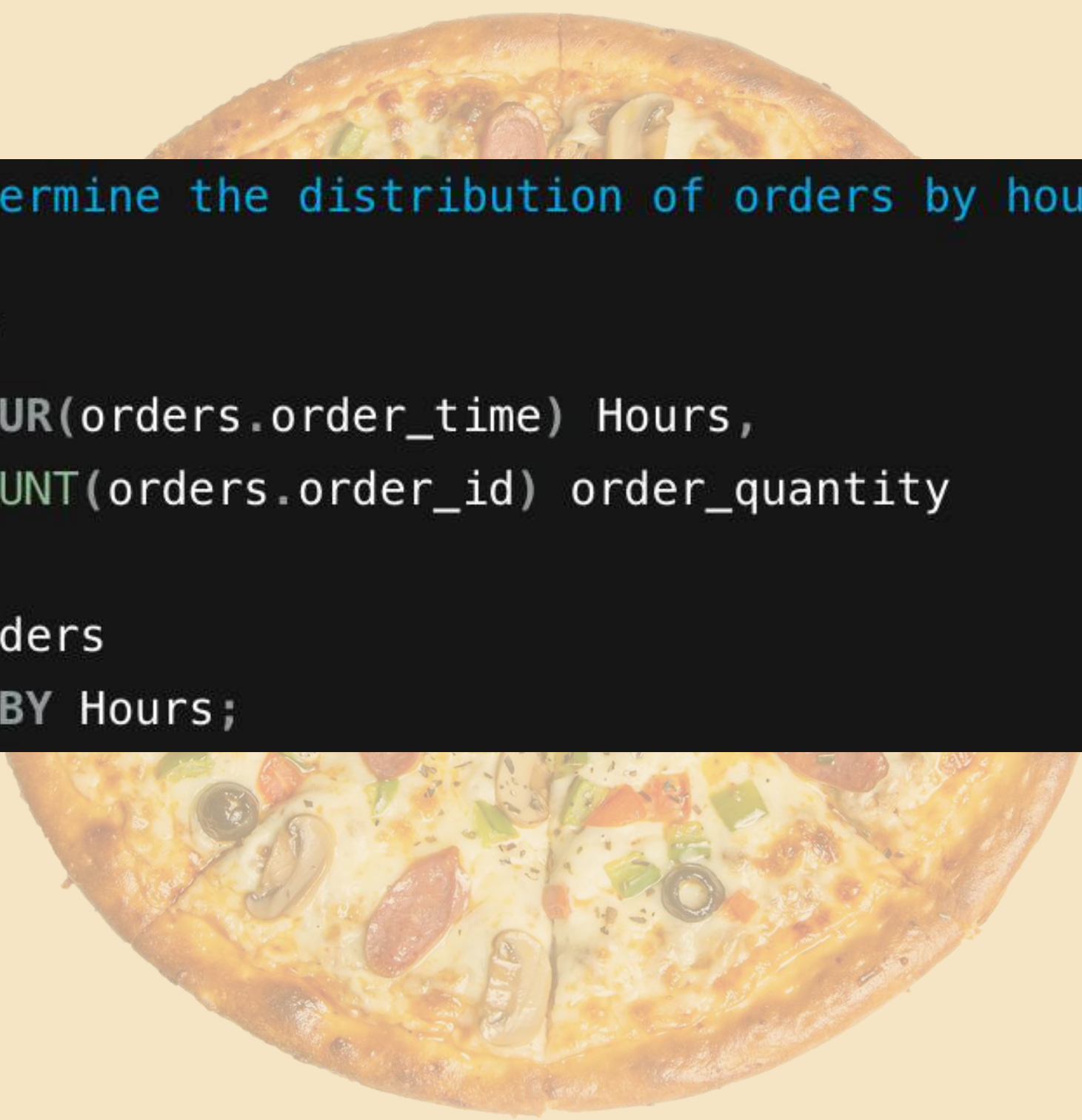


TOTAL QUANTITY OF EACH PIZZA CATEGORY



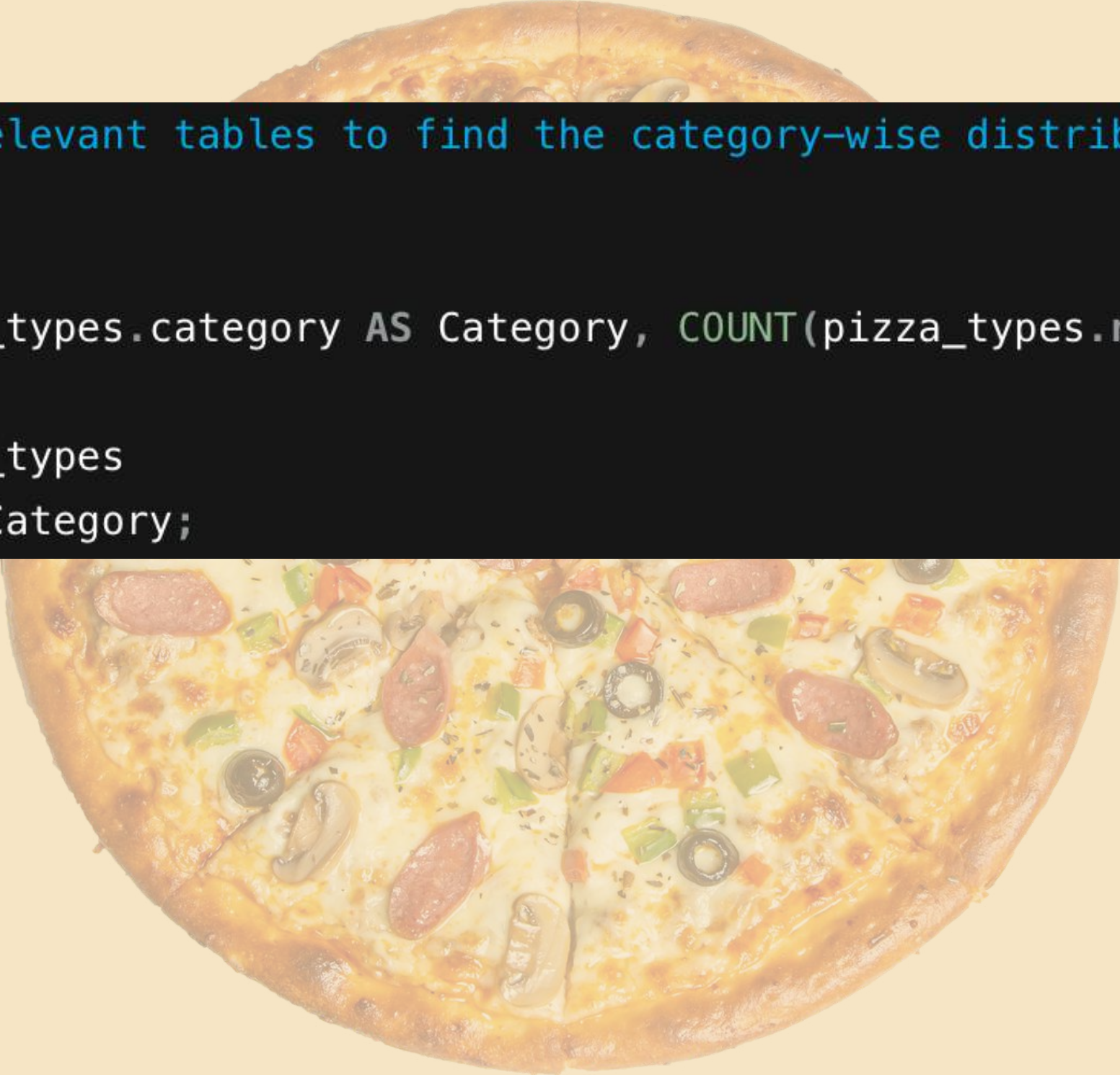
```
1  -- Join the necessary tables to find the total quantity of each pizza category ordered.
2
3  • SELECT
4      pizza_types.category,
5      SUM(order_details.quantity) AS total_quantity
6  FROM
7      pizza_types
8      JOIN
9      pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
10     JOIN
11     order_details ON order_details.pizza_id = pizzas.pizza_id
12 GROUP BY pizza_types.category
13 ORDER BY total_quantity DESC;
```


DISTRIBUTION OF ORDERS BY HOUR



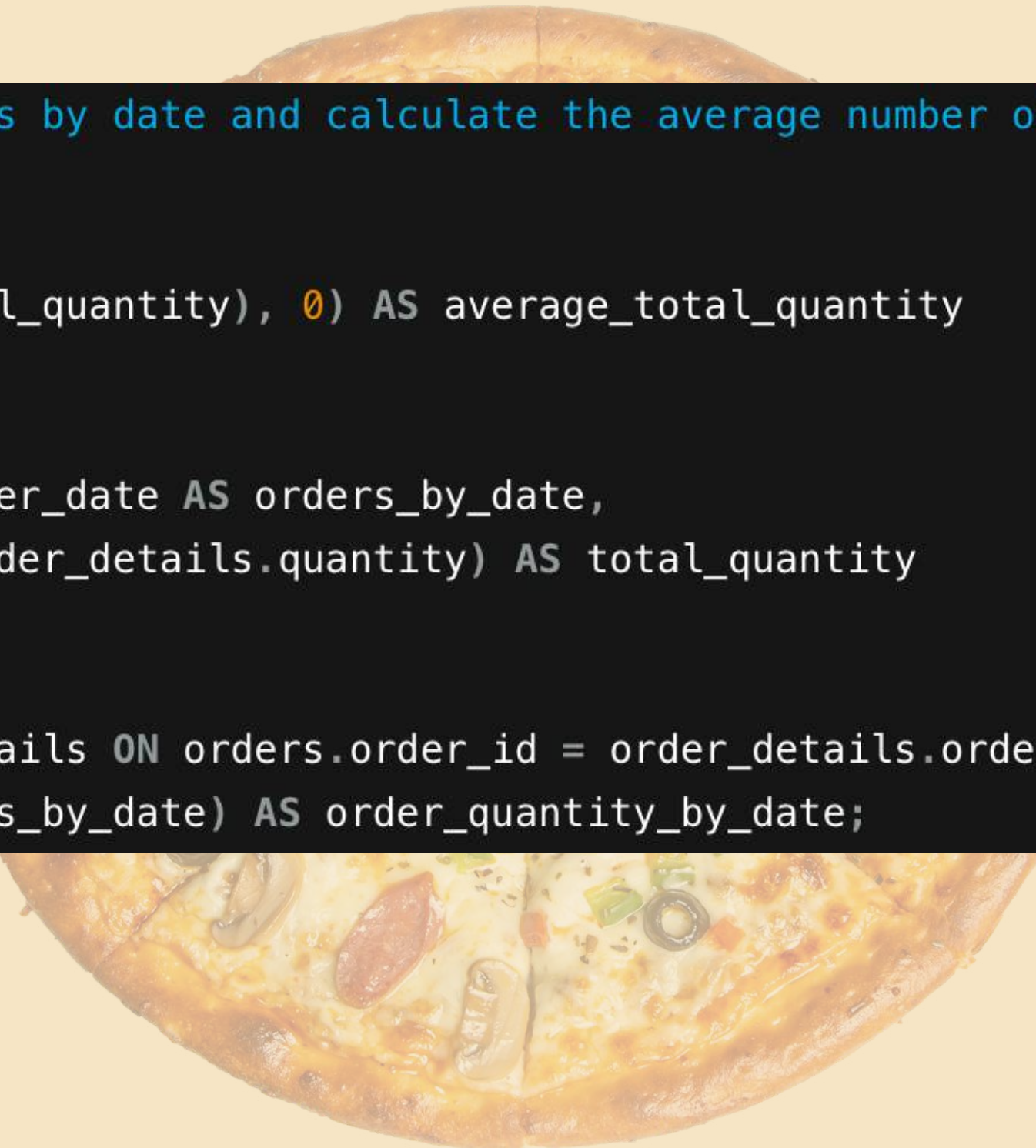
```
1  -- Determine the distribution of orders by hour of the day.
2
3  SELECT
4      HOUR(orders.order_time) Hours,
5      COUNT(orders.order_id) order_quantity
6  FROM
7      orders
8  GROUP BY Hours;
```

CATEGORY-WISE DISTRIBUTION



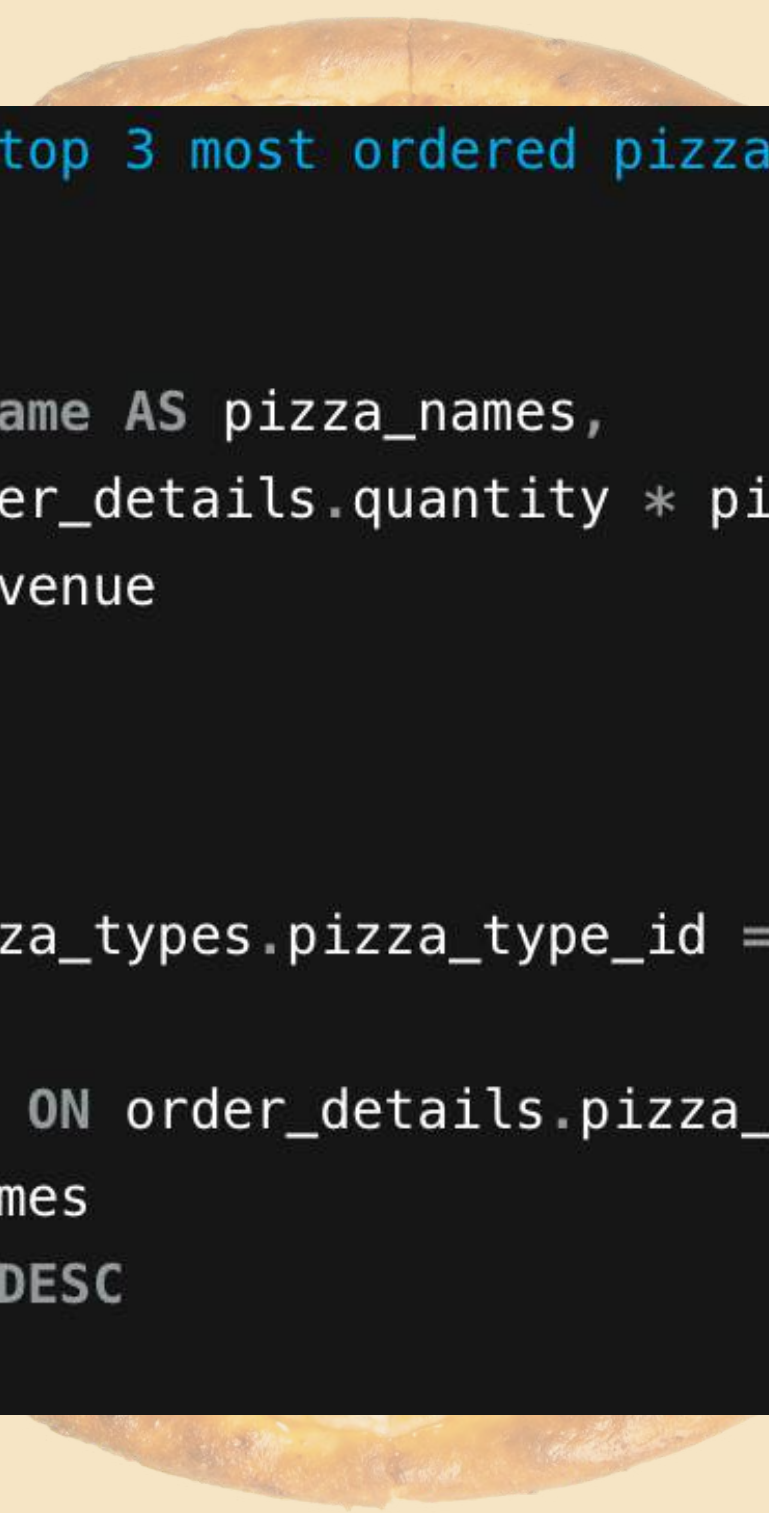
```
1  -- Join relevant tables to find the category-wise distribution of pizzas.
2
3  •  SELECT
4      pizza_types.category AS Category, COUNT(pizza_types.name)
5  FROM
6      pizza_types
7  GROUP BY Category;
```


AVERAGE NUMBER OF PIZZAS PER DAY



```
1  -- Group the orders by date and calculate the average number of pizzas ordered per day.
2
3  SELECT
4      ROUND(AVG(total_quantity), 0) AS average_total_quantity
5  FROM
6      (SELECT
7          orders.order_date AS orders_by_date,
8          SUM(order_details.quantity) AS total_quantity
9      FROM
10         orders
11        JOIN order_details ON orders.order_id = order_details.order_id
12       GROUP BY orders_by_date) AS order_quantity_by_date;
```

TOP 3 MOST ORDERED PIZZA BASED ON REVENUE



```
1  -- Determine the top 3 most ordered pizza types based on revenue.
2
3  * SELECT
4      pizza_types.name AS pizza_names,
5      ROUND(SUM(order_details.quantity * pizzas.price),
6             2) Revenue
7  FROM
8      pizza_types
9      JOIN
10     pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
11     JOIN
12     order_details ON order_details.pizza_id = pizzas.pizza_id
13  GROUP BY pizza_names
14  ORDER BY Revenue DESC
15  LIMIT 3;
```


ADVANCED QUESTIONS



PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE

```
1  -- Calculate the percentage contribution of each pizza type to total revenue.
2
3  •  SELECT
4      pizza_types.category,
5      ROUND(SUM(order_details.quantity * pizzas.price) / (SELECT
6          ROUND(SUM(order_details.quantity * pizzas.price),
7              2) AS total_sales
8          FROM
9              order_details
10             JOIN
11                 pizzas ON pizzas.pizza_id = order_details.pizza_id) * 100,
12          2) AS revenue
13  FROM
14      pizza_types
15      JOIN
16      pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
17      JOIN
18      order_details ON order_details.pizza_id = pizzas.pizza_id
19  GROUP BY pizza_types.category
20  ORDER BY revenue DESC;
```


CUMULATIVE REVENUE

```
1  -- Analyze the cumulative revenue generated over time.
2
3  * ⊖ WITH sales AS (
4      SELECT
5          orders.order_date,
6          SUM(order_details.quantity * pizzas.price) AS revenue
7      FROM order_details
8      JOIN pizzas
9          ON order_details.pizza_id = pizzas.pizza_id
10     JOIN orders
11         ON orders.order_id = order_details.order_id
12     GROUP BY orders.order_date
13 )
14 SELECT
15     order_date,
16     ⊖ ROUND(
17         SUM(revenue) OVER (ORDER BY order_date),
18         2
19     ) AS cumulative_revenue
20 FROM sales;
```

TOP 3 PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY

```
1      -- Determine the top 3 most ordered pizza types based on revenue for each pizza category.
2
3  * ⊖ WITH pizza_revenue AS (
4      SELECT
5          pizza_types.category,
6          pizza_types.name,
7          SUM(order_details.quantity * pizzas.price) AS revenue
8      FROM pizza_types
9      JOIN pizzas
10         ON pizza_types.pizza_type_id = pizzas.pizza_type_id
11      JOIN order_details
12         ON order_details.pizza_id = pizzas.pizza_id
13      GROUP BY
14          pizza_types.category,
15          pizza_types.name
16  )
17      SELECT
18          name,
19          revenue
20  ⊖ FROM (
21      SELECT
22          category,
23          name,
24          revenue,
25  ⊖      RANK() OVER (
26          PARTITION BY category
27          ORDER BY revenue DESC
28      ) AS rn
29      FROM pizza_revenue
30  ) ranked
31  WHERE rn <= 3;
```


EXCEL DASHBOARD



DATA PREPARATION

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	order_details_id	order_id	order_id_count	pizza_id	quantity	pizza_type_id	size	price	name	pizza_fl	category	ingredients	date	time	Revenue	
2	1	1	1	hawaiian_m	1	hawaiian	M	\$13.25	The Hawaiian Pizza	1	Classic	am, Pineapple, Mozzarella	01/01/15	11:38:36 AM	\$13.25	
3	11	6	0.5	bbq_ckn_s	1	bbq_ckn	S	\$12.75	The Barbecue Chicken Pizza	1	Chicken	s, Green Peppers, Tomato	01/01/15	12:29:36 PM	\$12.75	
4	71	27	0.333333333	bbq_ckn_l	1	bbq_ckn	L	\$20.75	The Barbecue Chicken Pizza	0	Chicken	s, Green Peppers, Tomato	01/01/15	3:11:17 PM	\$20.75	
5	90	36	0.5	bbq_ckn_m	1	bbq_ckn	M	\$16.75	The Barbecue Chicken Pizza	0	Chicken	s, Green Peppers, Tomato	01/01/15	4:54:09 PM	\$16.75	
6	63	24	0.25	bbq_ckn_m	1	bbq_ckn	M	\$16.75	The Barbecue Chicken Pizza	0	Chicken	s, Green Peppers, Tomato	01/01/15	2:23:01 PM	\$16.75	
7	12	6	0.5	the_greek_s	1	the_greek	S	\$12.00	The Greek Pizza	1	Classic	se, Tomatoes, Garlic, Bee	01/01/15	12:29:36 PM	\$12.00	
8	39	15	0.25	the_greek_s	1	the_greek	S	\$12.00	The Greek Pizza	0	Classic	se, Tomatoes, Garlic, Bee	01/01/15	1:33:00 PM	\$12.00	
9	2	2	0.2	classic_dlx_m	1	classic_dlx	M	\$16.00	The Classic Deluxe Pizza	1	Classic	rooms, Red Onions, Red	01/01/15	11:57:40 AM	\$16.00	
10	72	27	0.333333333	classic_dlx_m	1	classic_dlx	M	\$16.00	The Classic Deluxe Pizza	0	Classic	rooms, Red Onions, Red	01/01/15	3:11:17 PM	\$16.00	
11	91	36	0.5	classic_dlx_m	1	classic_dlx	M	\$16.00	The Classic Deluxe Pizza	0	Classic	rooms, Red Onions, Red	01/01/15	4:54:09 PM	\$16.00	
12	15	9	0.111111111	classic_dlx_s	1	classic_dlx	S	\$12.00	The Classic Deluxe Pizza	0	Classic	rooms, Red Onions, Red	01/01/15	12:52:01 PM	\$12.00	
13	3	2	0.2	five_cheese_l	1	five_cheese	L	\$18.50	The Five Cheese Pizza	1	Veggie	s, Smoked Gouda Cheese	01/01/15	11:57:40 AM	\$18.50	
14	37	15	0.25	five_cheese_l	1	five_cheese	L	\$18.50	The Five Cheese Pizza	0	Veggie	s, Smoked Gouda Cheese	01/01/15	1:33:00 PM	\$18.50	
15	4	2	0.2	ital_supr_l	1	ital_supr	L	\$20.75	The Italian Supreme Pizza	1	Supreme	collo, Tomatoes, Red Oni	01/01/15	11:57:40 AM	\$20.75	
16	7	3	0.5	ital_supr_m	1	ital_supr	M	\$16.50	The Italian Supreme Pizza	0	Supreme	collo, Tomatoes, Red Oni	01/01/15	12:12:28 PM	\$16.50	
17	9	4	1	ital_supr_m	1	ital_supr	M	\$16.50	The Italian Supreme Pizza	0	Supreme	collo, Tomatoes, Red Oni	01/01/15	12:16:31 PM	\$16.50	
18	10	5	1	ital_supr_m	1	ital_supr	M	\$16.50	The Italian Supreme Pizza	0	Supreme	collo, Tomatoes, Red Oni	01/01/15	12:21:30 PM	\$16.50	
19	18	9	0.111111111	ital_supr_l	1	ital_supr	L	\$20.75	The Italian Supreme Pizza	0	Supreme	collo, Tomatoes, Red Oni	01/01/15	12:52:01 PM	\$20.75	
20	19	9	0.111111111	ital_supr_s	1	ital_supr	S	\$12.50	The Italian Supreme Pizza	0	Supreme	collo, Tomatoes, Red Oni	01/01/15	12:52:01 PM	\$12.50	
21	6	2	0.2	thai_ckn_l	1	thai_ckn	L	\$20.75	The Thai Chicken Pizza	1	Chicken	tomatoes, Red Peppers, T	01/01/15	11:57:40 AM	\$20.75	
22	5	2	0.2	mexicana_m	1	mexicana	M	\$16.00	The Mexicana Pizza	1	Veggie	Peppers, Red Onions, Cil	01/01/15	11:57:40 AM	\$16.00	
23	20	9	0.111111111	mexicana_s	1	mexicana	S	\$12.00	The Mexicana Pizza	0	Veggie	Peppers, Red Onions, Cil	01/01/15	12:52:01 PM	\$12.00	
24	36	15	0.25	big_meat_s	1	big_meat	S	\$12.00	The Big Meat Pizza	1	Classic	eroni, Italian Sausage, Ch	01/01/15	1:33:00 PM	\$12.00	
25	8	3	0.5	prsc_argla_l	1	prsc_argla	L	\$20.75	The Prosciutto and Arugula Pizza	1	Supreme	San Daniele, Arugula, Moz	01/01/15	12:12:28 PM	\$20.75	
26	66	24	0.25	prsc_argla_s	1	prsc_argla	S	\$12.50	The Prosciutto and Arugula Pizza	0	Supreme	San Daniele, Arugula, Moz	01/01/15	2:23:01 PM	\$12.50	
27	64	24	0.25	ital_cpcllo_m	1	ital_cpcllo	M	\$16.00	The Italian Capocollo Pizza	1	Classic	pers, Tomatoes, Goat Che	01/01/15	2:23:01 PM	\$16.00	
28	17	9	0.111111111	ital_cpcllo_l	1	ital_cpcllo	L	\$20.50	The Italian Capocollo Pizza	0	Classic	pers, Tomatoes, Goat Che	01/01/15	12:52:01 PM	\$20.50	
29	73	27	0.333333333	spinach_fet_l	1	spinach_fet	L	\$20.25	The Spinach and Feta Pizza	1	Veggie	rooms, Red Onions, Feta	01/01/15	3:11:17 PM	\$20.25	
30	65	24	0.25	pepperoni_m	1	pepperoni	M	\$12.50	The Pepperoni Pizza	1	Classic	zzarella Cheese, Pepper	01/01/15	2:23:01 PM	\$12.50	
31	38	15	0.25	soppressata_l	1	soppressata	L	\$20.75	The Soppressata Pizza	1	Supreme	ina Cheese, Mozzarella Ch	01/01/15	1:33:00 PM	\$20.75	
32	13	7	1	spinach_supr_s	1	spinach_supr	S	\$12.50	The Spinach Supreme Pizza	1	Supreme	omatoes, Artichokes, Kal	01/01/15	12:50:37 PM	\$12.50	
33	14	8	1	spinach_supr_s	1	spinach_supr	S	\$12.50	The Spinach Supreme Pizza	0	Supreme	omatoes, Artichokes, Kal	01/01/15	12:51:37 PM	\$12.50	
34	16	9	0.111111111	green_garden_s	1	green_garden	S	\$12.00	The Green Garden Pizza	1	Veggie	oms, Tomatoes, Green Ol	01/01/15	12:52:01 PM	\$12.00	
35	21	9	0.111111111	spicy_ital_l	1	spicy_ital	L	\$20.75	The Spicy Italian Pizza	1	Supreme	Goat Cheese, Artichokes,	01/01/15	12:52:01 PM	\$20.75	
36	22	9	0.111111111	spin_pesto_l	1	spin_pesto	L	\$20.75	The Spinach Pesto Pizza	1	Veggie	matoes, Sun-dried Tomat	01/01/15	12:52:01 PM	\$20.75	
37	23	9	0.111111111	veggie_veg_s	1	veggie_veg	S	\$12.00	The Vegetables + Vegetables Pizza	1	Veggie	pers, Green Peppers, Red	01/01/15	12:52:01 PM	\$12.00	
38	45	17	0.1	five_cheese_l	1	five_cheese	L	\$18.50	The Five Cheese Pizza	0	Veggie	s, Smoked Gouda Cheese	01/01/15	1:53:00 PM	\$18.50	
39	43	17	0.1	bbq_ckn_l	1	bbq_ckn	L	\$20.75	The Barbecue Chicken Pizza	0	Chicken	s, Green Peppers, Tomato	01/01/15	1:53:00 PM	\$20.75	
40	47	17	0.1	ital_supr_m	1	ital_supr	M	\$16.50	The Italian Supreme Pizza	0	Supreme	collo, Tomatoes, Red Oni	01/01/15	1:53:00 PM	\$16.50	
41	50	17	0.1	mexicana_l	1	mexicana	L	\$20.25	The Mexicana Pizza	0	Veggie	Peppers, Red Onions, Cil	01/01/15	1:53:00 PM	\$20.25	
42	44	17	0.1	calabrese_m	1	calabrese	M	\$16.25	The Calabrese Pizza	1	Supreme	Tomatoes, Red Onions, F	01/01/15	1:53:00 PM	\$16.25	
43	46	17	0.1	four_cheese_m	1	four_cheese	M	\$14.75	The Four Cheese Pizza	1	Veggie	Cheese, Mozzarella Chees	01/01/15	1:53:00 PM	\$14.75	
44	48	17	0.1	ital_veggie_s	1	ital_veggie	S	\$12.75	The Italian Vegetables Pizza	1	Veggie	natoes, Zucchini, Red Pep	01/01/15	1:53:00 PM	\$12.75	
45	49	17	0.1	mediterraneo_m	2	mediterraneo	M	\$16.00	The Mediterranean Pizza	1	Veggie	Sun-dried Tomatoes, Feta	01/01/15	1:53:00 PM	\$32.00	
46	51	17	0.1	peppr_salami_s	1	peppr_salami	S	\$12.50	The Pepper Salami Pizza	1	Supreme	llo, Pepperoni, Tomatoes	01/01/15	1:53:00 PM	\$12.50	
47	52	17	0.1	spinach_fet_l	1	spinach_fet	L	\$20.25	The Spinach and Feta Pizza	0	Veggie	rooms, Red Onions, Feta	01/01/15	1:53:00 PM	\$20.25	
◀ ▶ Pizza_sales_merged Pivot Dashboard +																

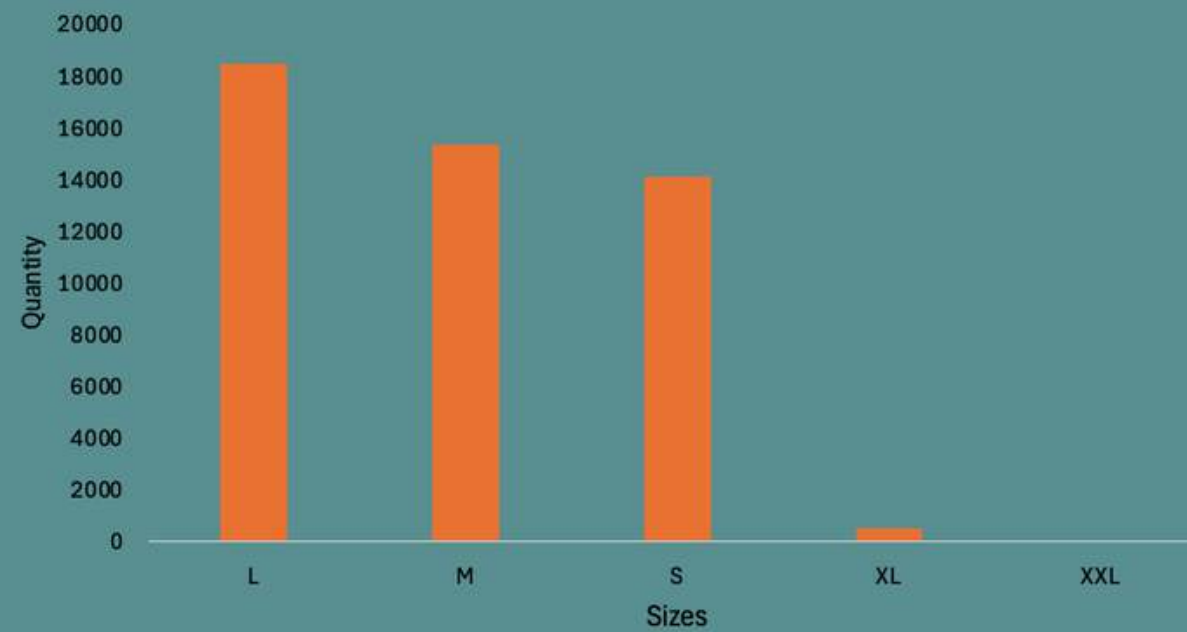
PIVOT TABLES

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1					Sizes	Count of sizes			Categories	Distribution		Categories	Sum of Revenue			Categories	Sum of Revenue		
2					L	18526			Chicken	6		01/01/15	\$2,713.85			Chicken	\$1,27,611.75		
3	Total number of orders placed				M	15385			Classic	8		02/01/15	\$5,445.75			The Thai Chicken Pizza	\$43,434.25		
4	21350				S	14137			Supreme	9		03/01/15	\$8,108.15			The Barbecue Chicken Pizza	\$42,768.00		
5					XL	544			Veggie	9		04/01/15	\$9,863.60			The California Chicken Pizza	\$41,409.50		
6					XXL	28						05/01/15	\$11,929.55			Classic	\$1,00,615.50		
7												06/01/15	\$14,358.50			The Classic Deluxe Pizza	\$38,180.50		
8												07/01/15	\$16,560.70			The Hawaiian Pizza	\$32,273.25		
9	Total revenue generated								Categories	Sum of quantity		08/01/15	\$19,399.05			The Pepperoni Pizza	\$30,161.75		
10	\$8,17,860.05				Sizes	Sum of quantity			01-Jan	162		09/01/15	\$21,526.40			Supreme	\$99,248.50		
11					The Classic Deluxe Pizza	2453			02-Jan	165		10/01/15	\$23,990.35			The Spicy Italian Pizza	\$34,831.25		
12					The Barbecue Chicken Pizza	2432			03-Jan	158		11/01/15	\$25,862.65			The Italian Supreme Pizza	\$33,476.75		
13					The Hawaiian Pizza	2422			04-Jan	106		12/01/15	\$27,781.70			The Sicilian Pizza	\$30,940.50		
14	Row Labels	Max. of price			The Pepperoni Pizza	2418			05-Jan	125		13/01/15	\$29,831.30			Veggie	\$85,112.95		
15	The Greek Pizza	\$35.95			The Thai Chicken Pizza	2371			06-Jan	147		14/01/15	\$32,358.70			The Four Cheese Pizza	\$32,265.70		
16	The Brie Carre Pizza	\$23.65							07-Jan	138		15/01/15	\$34,343.50			The Mexicana Pizza	\$26,780.75		
17	The Italian Vegetables Pizza	\$21.00							08-Jan	173		16/01/15	\$36,937.65			The Five Cheese Pizza	\$26,066.50		
18	The Soppressata Pizza	\$20.75							09-Jan	127		17/01/15	\$39,001.75						
19	The Spinach Supreme Pizza	\$20.75			Row Labels	Sum of quantity			10-Jan	146		18/01/15	\$40,978.60						
20	The Spicy Italian Pizza	\$20.75			Classic	14888			11-Jan	116		19/01/15	\$43,365.75						
21	The Chicken Pesto Pizza	\$20.75			Supreme	11987			12-Jan	119		20/01/15	\$45,763.65						
22	The Chicken Alfredo Pizza	\$20.75			Veggie	11649			13-Jan	120		21/01/15	\$47,804.20						
23	The California Chicken Pizza	\$20.75			Chicken	11050			14-Jan	150		22/01/15	\$50,300.90						
24	The Southwest Chicken Pizza	\$20.75							15-Jan	123		23/01/15	\$52,724.60						
25	The Italian Supreme Pizza	\$20.75							16-Jan	158		24/01/15	\$55,013.85						
26	The Spinach Pesto Pizza	\$20.75							17-Jan	125		25/01/15	\$56,631.40						
27	The Barbecue Chicken Pizza	\$20.75			Hour	Order quantity			18-Jan	122		26/01/15	\$58,515.80						
28	The Thai Chicken Pizza	\$20.75			12 PM	2520			19-Jan	142		27/01/15	\$61,043.85						
29	The Pepper Salami Pizza	\$20.75			1 PM	2455			20-Jan	143		28/01/15	\$63,059.85						
30	The Prosciutto and Arugula Pizza	\$20.75			6 PM	2399			21-Jan	129		29/01/15	\$65,105.15						
31					5 PM	2336			22-Jan	158		30/01/15	\$67,375.45						
32					7 PM	2009			23-Jan	152		31/01/15	\$69,793.30						
33					4 PM	1920			24-Jan	142		01/02/15	\$72,982.50						
34	Pizza Names	Total Revenue			8 PM	1642			25-Jan	102		02/02/15	\$75,311.10						
35	The Thai Chicken Pizza	\$43,434.25			2 PM	1472			26-Jan	113		03/02/15	\$77,925.90						
36	The Barbecue Chicken Pizza	\$42,768.00			3 PM	1468			27-Jan	151		04/02/15	\$80,159.80						
37	The California Chicken Pizza	\$41,409.50			11 AM	1231			28-Jan	118		05/02/15	\$82,375.60						
38					9 PM	1198			29-Jan	119		06/02/15	\$84,885.55						
39					10 PM	663			30-Jan	138		07/02/15	\$87,123.20						
40					11 PM	28			31-Jan	145		08/02/15	\$89,158.20						
41					10 AM	8			01-Feb	191		09/02/15	\$91,353.55						
42	Pizza Names	Total Revenue																	
	Pizza_sales_merged	Pivot	Dashboard	+															

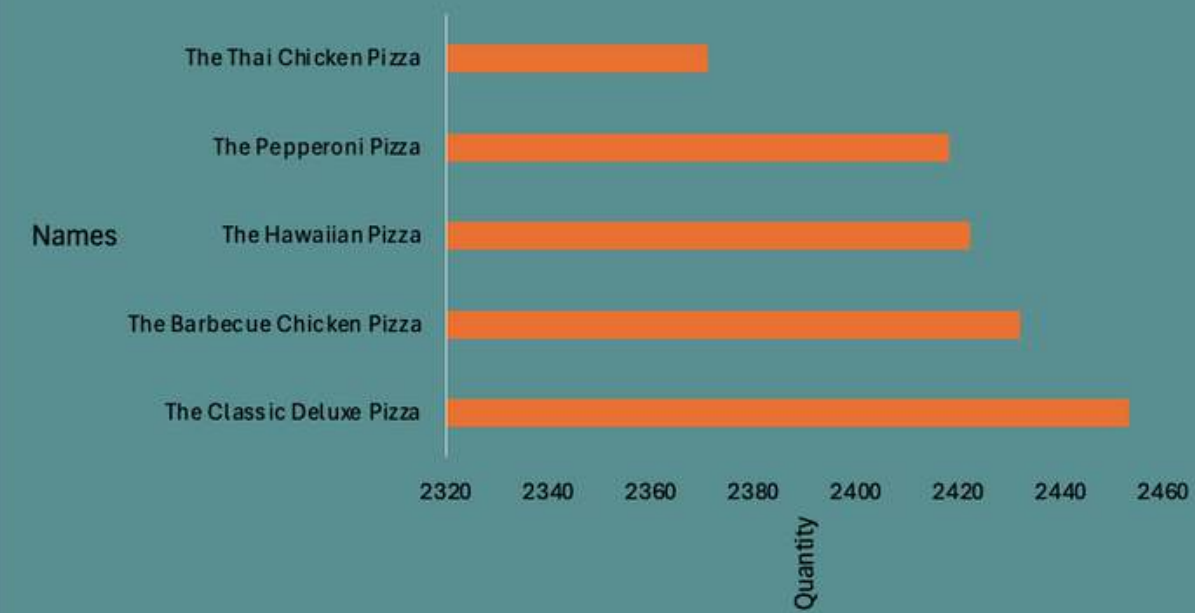
DASHBOARD

PIZZA SALES ANALYSIS

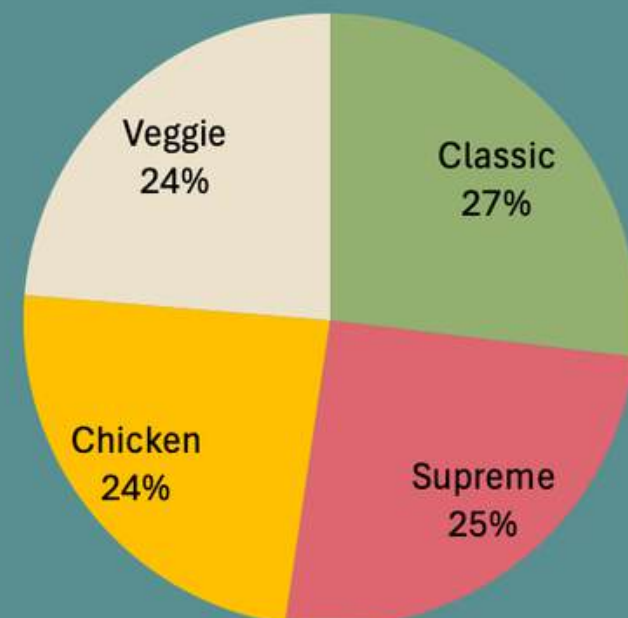
Most common pizza size ordered



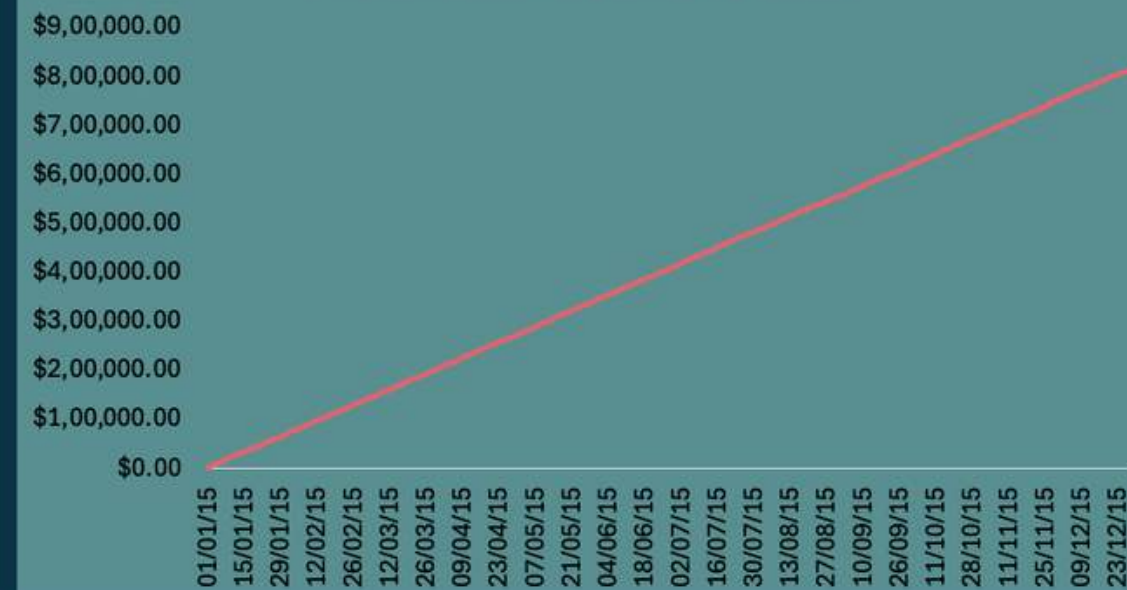
Top 5 most ordered pizza



Percentage contribution of each pizza type to total revenue



Cumulative revenue over time



THANK
YOU

